

Global PM2.5 Pollution Analysis Using Data Mining Techniques: A CRISP-DM Approach

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Abstract—On a worldwide scale, PM2.5 pollution raises significant environmental and health issues that necessitate thorough analysis and prediction. Applying the SatPM2.5 dataset, this paper employs data mining approaches to analyse the trends of PM2.5 worldwide between 1998 and 2023 and evaluates the predictive models in forecasting future values. Following the CRISP-DM framework, two machine learning models are employed in the data mining process, including data preprocessing, feature extraction, clustering, and prediction. To test regional differences, an additional dataset of cities is applied.

Earlier studies have indicated that hybrid and ensemble machine learning algorithms outperform traditional algorithms in predicting PM2.5 [1]–[4]. However, the current paper proves that simple approaches can prove equally efficient when applied to large-scale datasets with significant time dependency. It has been proven from experiments that PM2.5 exhibits significant time dependency, and using lagged features helps achieve high prediction performance. Linear regression model demonstrated the best results (RMSE = 2.66 $\mu\text{g}/\text{m}^3$, $R^2 = 0.96$), outperforming Random Forest.

The findings of the clustering study indicate that there is a disparity between countries when it comes to the quality of air by showing different regional patterns of pollution. These results show that long-term trends are responsible for changes in PM2.5 levels instead of weather phenomena. Thus, it can be seen that temporal modelling and clustering provide efficient environmental data analysis. To increase the accuracy and precision of the model, further studies should take into account meteorological and satellite information.

Keywords—PM2.5, Air Pollution, Data Mining, Machine Learning, CRISP-DM, Clustering, Time-Series Forecasting

I. INTRODUCTION

Fine Particulate Matter (PM2.5), being capable of penetrating deep into the lungs and contributing to various diseases related to the heart, lungs, and brain, renders it one of the most hazardous pollutants [16]. The PM2.5 burden remains unequal across different regions of the world, with its concentration being relatively higher in developing countries [3]. As such, it becomes critical to be able to understand the past and predict the future trends in PM2.5 concentration.

Machine learning has evolved into a powerful tool for PM2.5 prediction that works well in diverse datasets across multiple geographical scales [1], [2], [4]. By taking into account complex spatial-temporal relationships, sophisticated

machine learning techniques such as deep learning and model hybrids have significantly boosted the predictive power of such algorithms [6], [8]. Furthermore, accurate high-resolution estimates of PM2.5 levels on a global scale have been enabled through the use of ensembles and satellite data [3], [7], [9], [10], and [15].

Although much progress has been made to date, many recent research works involve highly complicated models with significant amounts of meteorological and satellite data. In contrast, this paper investigates the possibility of predicting the long-term behaviours of PM2.5 levels solely based on historical observations through a simpler model formulation.

The major contributions of this research paper are as follows:

- The analysis of PM2.5 patterns for a period of 25 years globally.
- Demonstrating that temporal models based on basic techniques yield very high accuracy in terms of forecasting.
- The use of predictive and clustering analysis together in order to gain better insights into the patterns.
- Verification through independent environmental datasets from city level.

The main research questions answered in this paper are as follow:

- What kinds of temporal and spatial patterns can be observed PM2.5 data globally in this long term?
- Can we forecast PM2.5 accurately using only historical information?
- How can nations be classified in terms of their long-term pollution trends?

CRISP-DM methodology is utilized in order to conduct this analysis systematically.

II. RELATED WORK

Machine learning approaches to predict PM2.5 levels have been extensively studied in the most recent literature. As described in the work of Patel et al. [1], since hybrid and ensemble models can account for complex nonlinear relationships, they consistently outperform individual models. Gradient boosting approaches provide accurate predictions in

the case of PM2.5 estimates, as noted by Makhdoomi et al. [2].

Deep ensemble learning algorithms are used by Yu et al. [3] to estimate PM2.5 levels worldwide, showing distant exposure differences and achieving high predictive accuracy. In addition, Panda et al. [4] show that when dealing with tabular air pollution data sets, tree-based ensemble models yield higher accuracy than deep learning models.

In Adhikari and Hussain [5], the concept of machine learning is embedded into an environmental context that transcends the idea of prediction. In this respect, the roles of machine learning in sensing, forecasting, and mitigation methods are emphasised, while strong results in modelling of spatiotemporal dependencies have been obtained through deep learning models, for instance, CNN-LSTM models [6].

Considerable success was achieved in the field of satellite PM2.5 estimates, as summarised in Lee [7] and Ahang et al. [15]. Hybrid machine learning models can be used to achieve better forecasting results [8]. Furthermore, geographically weighted and ensemble models can enhance the resolution and interpretability of the results [9, 10].

Patterns of pollution have extensively relied upon the use of clustering techniques. Whereas Unik et al. [14] emphasise the importance of incorporating varied sources of data for effective models, Govender and Sivakumar [13] point out the efficiency of the K-Means clustering technique in discovering pollution regimes.

As this paper emphasises a simple and understandable strategy using clustering for temporal pattern discovery, earlier works have focused more on complex modelling techniques or regional data sets.

III. METHODOLOGY (CRISP-DM)

As a result of using the CRISP-DM framework for this research, an approach to data mining is provided in a structured and repeatable way.

A. Business Understanding

To analyse global PM2.5 trends and develop prediction models that could reveal any temporal dependencies is the aim.

B. Data Understanding

The SatPM2.5 dataset includes 6318 observations for 242 regions over the time period ranging from 1998 to 2023. Even though all results were available and there were no missing values, 26 values for PM2.5 were excluded because of their physical inconsistency.

C. Data Preprocessing

The data preprocessing entailed:

- Removal of any duplicates ensure data integrity.
- Selection of relevant features (Year, Region, PM2.5)

- Sorting by the region and year to ensure a chronological sequence of observation.

Capturing temporal dependencies entailed feature engineering:

- PM2.5_lag1 (previous year)
- PM2.5_lag2 (two years ago)

5808 observations in the end dataset

D. Modelling

Two models were implemented:

- Linear Regression (Interpretable baseline)
- Random Forest (nonlinear ensemble model)

Train-test split:

- Training: 2000-2018...
- Testing: 2019-2023

E. Evaluation

Models were evaluated using:

- Root Mean Square Error (RMSE)
- Coefficient of Determination (R^2)

IV. DATA PREPARATION

In order to ensure that the data is physically valid, all PM2.5 values less than zeros have been removed. The use of lag-based features has made it possible to incorporate temporal dependencies into the model.

The following equation represents the relationship between input variables and the target variable:

$$PM25_t = f(PM25_{t-1}, PM25_{t-2}, Year)$$

Using this technique, it becomes possible to incorporate both short-term and long-term temporal dependencies.

V. RESULTS AND EVALUATION

A. Predictive Performance

So.No	Performance		
	Model	RMSE	R^2
1.	Linear Regression	2.66	0.96
2.	Random Forest	3.06	0.95

The trends of PM2.5 exhibit excellent linear temporal properties as reflected in the effectiveness of the Linear Regression model compared to the Random Forest model.

Model accuracy is shown through the similarity of the predicted and observed figures, as illustrated in Fig 1. Prediction accuracy is reflected in the slight discrepancy between the estimated and observed figures.

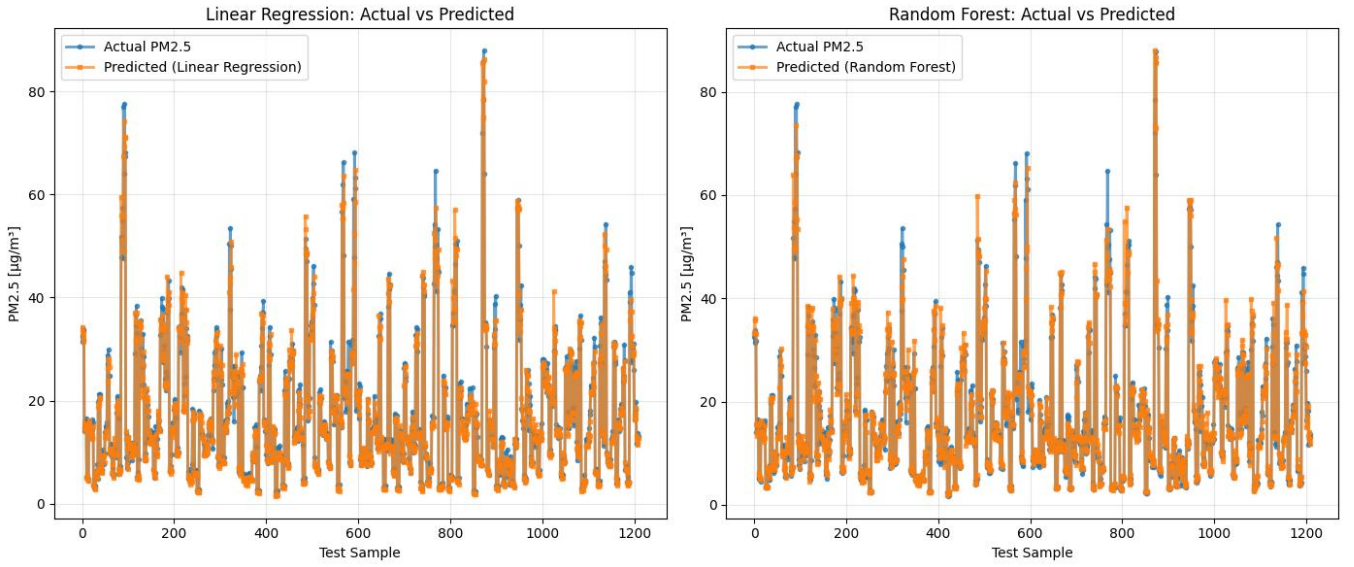


Fig. 1. Actual vs Predicted PM2.5 values.

B. Interpretation

The model accounts for about 96 per cent of the variance in PM2.5 concentration values because of the high R-squared value of 0.96. The importance of the lag variable confirms the influence of past periods on PM2.5 concentration.

The finding is in line with earlier studies on the time-dependency of environmental pollutants data [1], [12].

VI. CLUSTERING ANALYSIS

In order to uncover trends in PM2.5 concentration among countries, the K-means clustering approach was employed.

Four (K=4) clusters were selected via the Elbow Method.

The clusters are as follows:

- Cluster 1: Regions characterised by severe pollution (East/South Asia)
- Cluster 2: Moderately polluted regions (parts of Africa/the Middle East)
- Cluster 3: Least polluted region (Europe/island countries)
- Cluster 4: Regions showing fluctuations and/or improvement

The clusters exhibit definite historical trends, which indicate regional differences in terms of economic development, population size, and legislation policies, as illustrated in Fig 2.

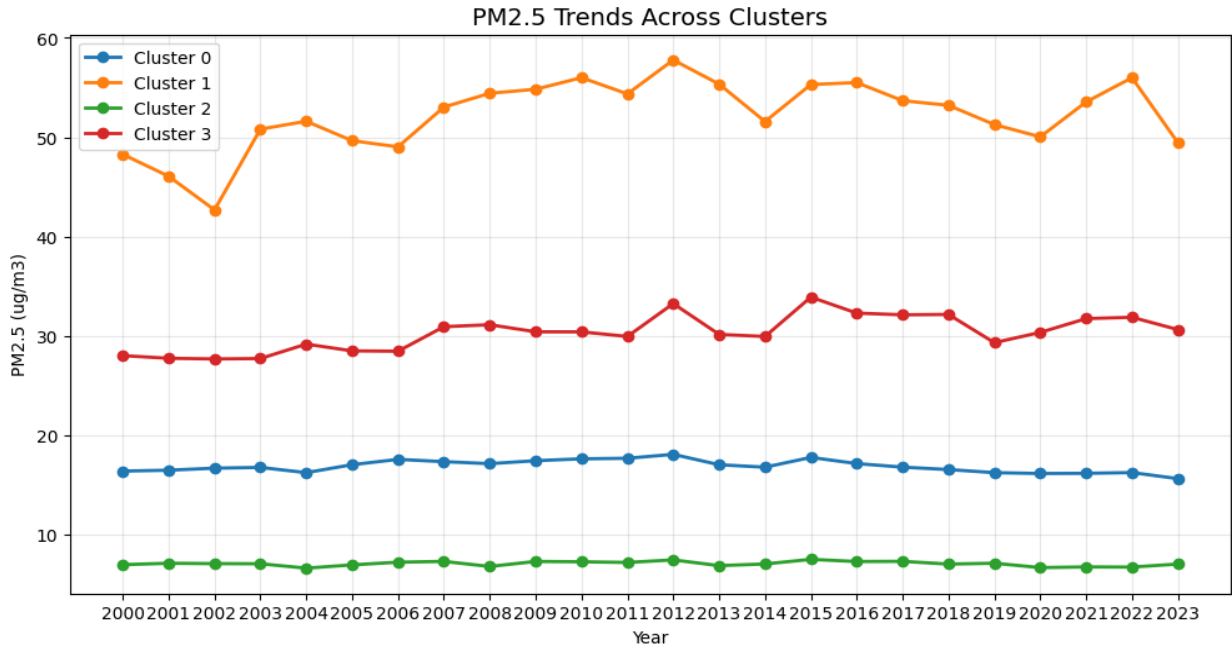


Fig 2: Cluster Trends Plot

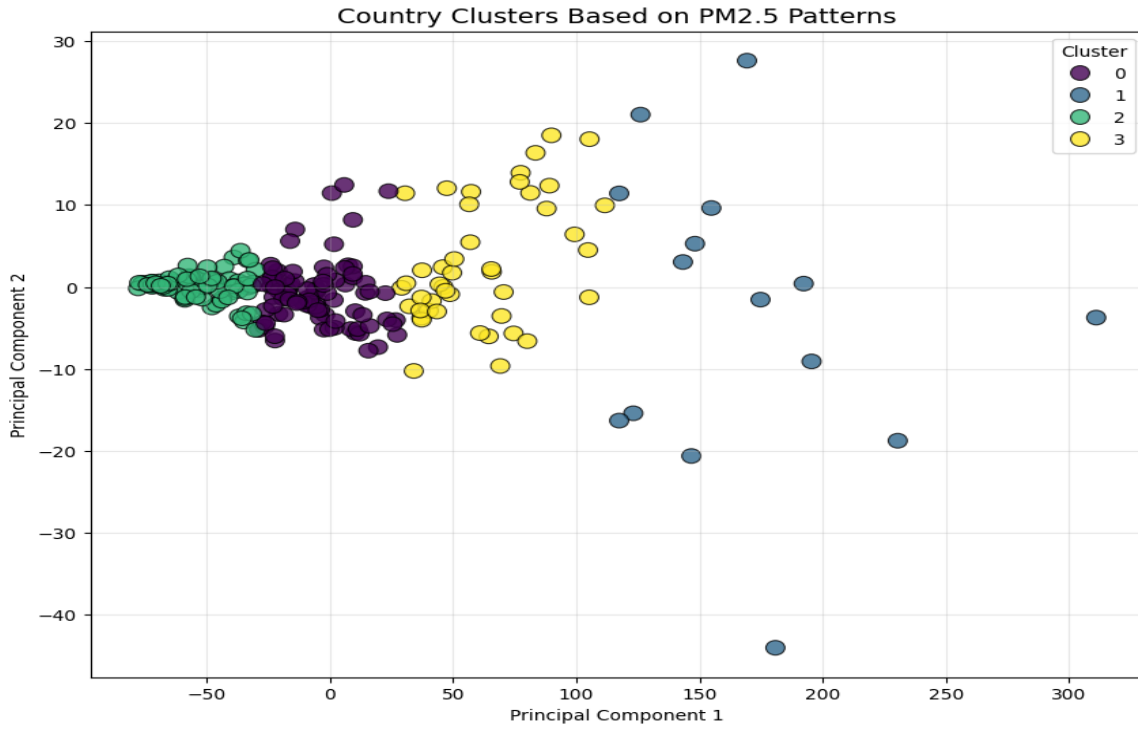


Fig 3: PCA Cluster Visualisation

This information, which reveals the non-uniform distribution of pollutants in different regions, confirms the trend of inequality revealed in [3].

VII. VALIDATION USING CITY-LEVEL DATA

To test the assumptions of the model, the study considered the variations of $Om_{2.5}$ at the city-level based on an additional dataset.

The outcome of the correlation analysis indicated the following:

- Temperature correlation is negligible (0.05).
- Humidity correlation is negatively moderated (~ 0.21).

From this finding, it can be inferred that compared to long-term historical trends, short-term weather changes have minimal impact on $PM_{2.5}$.

Therefore, the results support the idea that temporal factors could be employed to predict $PM_{2.5}$ levels.

VIII. DISCUSSION

Results from the current study give new insights and are coherent with earlier studies. Temporal dependence plays an important role in predicting $PM_{2.5}$ concentrations, which agrees with findings reported in [1] and [2]. However, unlike other papers that stress complex models [6], [8], the current study proves that simple models give similar results when there is adequate temporal dependency.

Cluster analysis gives similar results to those found in [3], pointing out the disparities of pollution concentration on a global scale. A more comprehensive understanding of $PM_{2.5}$ behaviour can be gained by combining cluster analysis and forecasting.

IX. CONCLUSION AND FUTURE WORK

In this study, data mining approaches are used for a complete $PM_{2.5}$ analysis around the globe. Trends for $PM_{2.5}$ analysis depend highly on time, as proved by the superior prediction capability of linear regression.

Clustering exposed differences in air quality around the world, creating interesting regional patterns. Major reasons behind $PM_{2.5}$ variations are historic trends, based on the validation results.

Future work will focus on:

- Incorporating satellite data and meteorology [3], [7], [15]
- Ensemble and hybrid model analyses [1], [8], [9]
- Spatial prediction models with high resolution [10]

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Project Files: <https://drive.google.com/drive/folders/1zcociaTaLBl10VgTu0m1FdR7MVTmZrg?usp=sharing>

Github Link: <https://github.com/Lakshmi-Tejaswi-04/Global-PM25-Analysis>